EAST WEST UNIVERSITY

Course Details

Course Title: Data Communications

Couse Code: CSE 350

Section: 01

Submitted To

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Star Topology:

Star Topology is a network setup in which each device is connected to a central node called a hub (switch). The hub manages the data flow between the devices. In this topology, all the nodes in a star topology are connected to the Hub and the Hub is responsible for the transmission of the data in the network. Whenever any node wants to transmit data to another node, it first transmits data to the central node (hub), and then it transfers the data to all the nodes on the network. After receiving the data, the hub checks for the destination address, and if the address matches the data is accepted otherwise data is rejected. The figure below shows the star topology.

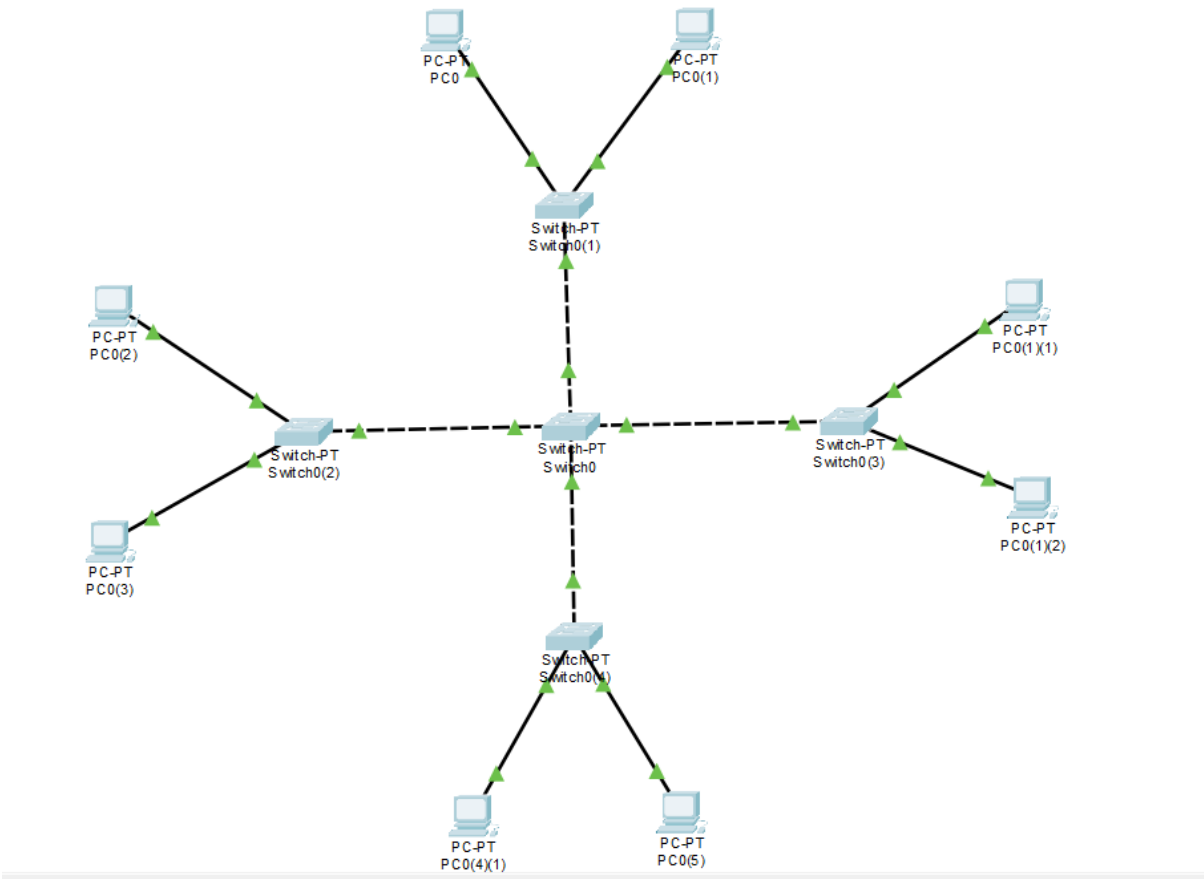

Figure: Star Topology

Mesh Topology

A mesh topology is a network setup where each computer and network device is interconnected with one another. This design provides high redundancy and fault tolerance, as data can take

multiple paths to reach its destination if one link fails. Mesh topology can be full (every device connected to every other device) or partial (devices connected to only some others), making it reliable but often more expensive and complex due to the large number of connections and cables required.

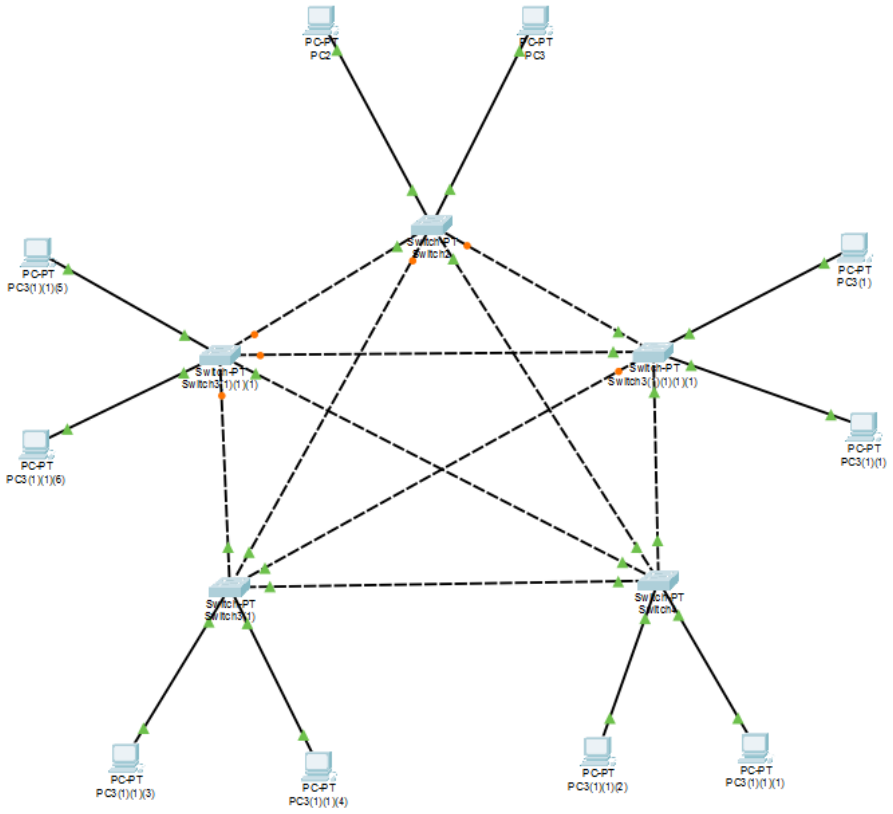

Figure: Mesh Topology

Hybrid Topology

A Hybrid Topology is any network that is created by connecting two or more different fundamental topologies (like Star, Bus, Ring) to form a larger, more robust network. A hybrid topology that combines Star and Mesh structures achieves a balance between centralized management and high fault tolerance.

In a Star Mesh hybrid topology, a Star network segment is connected to a Mesh topology where the Mesh core, consisting of critical devices like servers or main switches, provides high redundancy and fault tolerance, ensuring uninterrupted data flow. Each Star segment connects its devices to a central switch, which is then linked to the Mesh. This setup allows efficient local access through the Star, while the Mesh ensures reliability for the core. Failures in a Star segment affect only that segment, and new Star networks can be easily added, though careful planning and higher costs are involved due to the Mesh backbone's complexity.

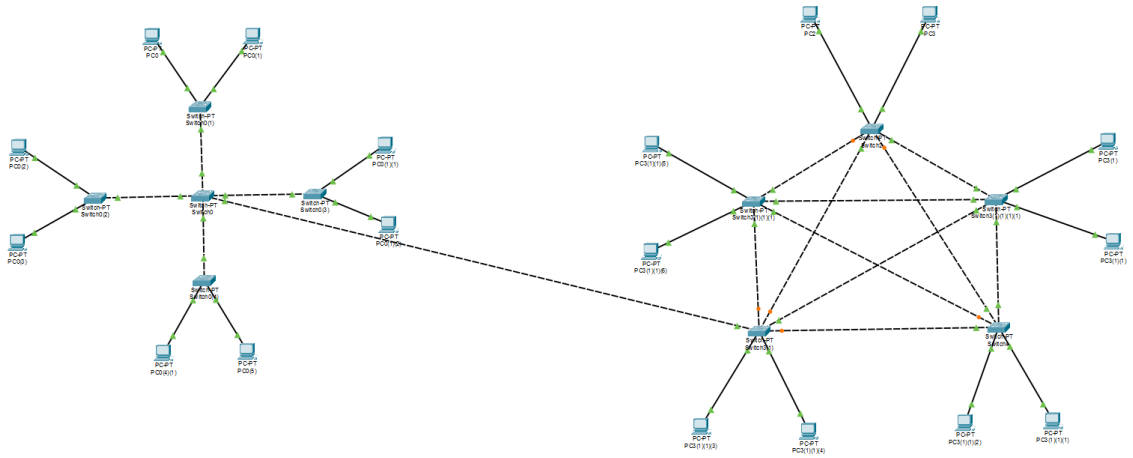

Figure: Hybrid topology using Star & Mesh topology.